

# EEN1095 Bi-weekly Log of GenAI Interactions

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| <b>Student Name</b>          | Darshan Mahadeva Naika                                                                                                  | <b>Student ID</b>   | A00090581      |
| <b>Project Title</b>         | Design and Evaluation of Low-Sensitivity Digital Filters Implemented in Python Under Varying Numerical Precision Levels | <b>Supervisor</b>   | Martin Collier |
| <b>Implementation Period</b> | Week 3 - Week 4                                                                                                         | <b>Version Date</b> | 05/06/2026     |

**Guidance:** Maintain this as a single cumulative document throughout EEN1095. Add a new section every two weeks and do not remove earlier records. All significant GenAI interactions must be logged. Each entry should record the tool used, the prompts submitted, the outputs generated, and a brief critical reflection on how the interaction was used, including any limitations. In addition, every two weeks you are required to upload to Loop a file containing the full record of your GenAI interactions for that period. This log in conjunction with the upload will allow you to summarize your interactions according to the [TOPIC](#) format recommended in the Guidelines for the Use of GenAI in Masters Projects. If there were no GenAI interactions during a reporting period, please write **None** in the relevant section.

| GenAI Interaction Week 1-2                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                             |
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| <b>Tool Name and model and date used</b>                                                                                                                                                                                                | Claude (Anthropic) — Claude Sonnet — Used throughout Week 1-2 (11/05/2026 — 22/05/2026)                                                                                                                                                                                                                                                     |
| <b>Output, Prompts, and Iterations</b><br><br><i>Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop</i> | Prompts submitted to Claude during Week 1-2 covered the following activities: Clarification of technical concepts — prompts asking Claude to explain digital ladder filter theory (Bruton 1975), wave digital filter theory (Fettweis 1986), and the mathematical definitions of pole displacement, stability margin, and sensitivity norm. |
| <b>Critical Analysis</b><br><br><i>Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project</i>                                                                      | Benefits: Claude was highly effective for explaining technical DSP concepts in clear language with the project content. It helped clarify mathematical definitions from the referenced academic papers. GenAI was used as a productivity and clarification tool — not as a source of original research results or experimental data.        |

| GenAI Interaction Week 3-4                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                            |
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| <b>Tool Name and model and date used</b>                                                                                                                                                                                                | Claude (Anthropic) — Claude Sonnet — Used throughout Week 3-4 (23/05/2026 — 05/06/2026)                                                                                                                                                                                                                                                    |
| <b>Output, Prompts, and Iterations</b><br><br><i>Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop</i> | Prompts submitted to Claude during Week 1-2 covered the following activities: Clarification of technical concepts — prompts asking Claude to explain digital ladder filter theory (Bruton 1975), wave digital filter theory (Fettweis 1986), and the mathematical definitions of pole displacement, stability margin, and sensitivity norm |
| <b>Critical Analysis</b><br><br><i>Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project</i>                                                                      | Benefits: Claude was highly effective for explaining technical DSP concepts in clear language with the project content. It helped clarify mathematical definitions from the referenced academic papers. GenAI was used as a productivity and clarification tool — not as a source of original research results or experimental data.       |

| GenAI Interaction Week 5-6                                                                                                                                                                                                              |  |
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| <b>Tool Name and model and date used</b>                                                                                                                                                                                                |  |
| <b>Output, Prompts, and Iterations</b><br><br><i>Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop</i> |  |
| <b>Critical Analysis</b><br><br><i>Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project</i>                                                                      |  |

| GenAI Interaction Week 7-8                                                                                                                                                                                    |  |
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| <b>Tool Name and model and date used</b>                                                                                                                                                                      |  |
| <b>Output, Prompts, and Iterations</b><br><br><i>Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of</i> |  |

|                                                                                                                                                                    |  |
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| <i>the file uploaded to Loop</i>                                                                                                                                   |  |
| <b>Critical Analysis</b><br><br><i>Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project</i> |  |

| GenAI Interaction Week 9-10                                                                                                                                                                                                             |  |
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| <b>Tool Name and model and date used</b>                                                                                                                                                                                                |  |
| <b>Output, Prompts, and Iterations</b><br><br><i>Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop</i> |  |
| <b>Critical Analysis</b><br><br><i>Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project</i>                                                                      |  |

| GenAI Interaction Week 11-12                                                                                                                                                                                                            |  |
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| <b>Tool Name and model and date used</b>                                                                                                                                                                                                |  |
| <b>Output, Prompts, and Iterations</b><br><br><i>Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop</i> |  |
| <b>Critical Analysis</b><br><br><i>Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project</i>                                                                      |  |

| GenAI Interaction Week 13-14                                                                                                                                                                                                            |  |
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| <b>Tool Name and model and date used</b>                                                                                                                                                                                                |  |
| <b>Output, Prompts, and Iterations</b><br><br><i>Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop</i> |  |
| <b>Critical Analysis</b><br><br><i>Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project</i>                                                                      |  |